CSE STUDY HUB

Digital Forensics & Incident Response

UNIT 5 · AI-Based Digital Forensics and Incident Response

What this unit is about

This unit builds understanding of **AI-Based Digital Forensics and Incident Response**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. AI/ML use in forensic triage, classification and anomaly detection

2. Supervised vs unsupervised learning for security evidence

3. Behaviour analytics and detection of suspicious patterns

4. Automated log correlation, alert prioritisation and threat intelligence

5. Benefits: speed, scale and pattern discovery

6. Risks: false positives/negatives, bias, explainability and adversarial inputs

7. Human validation, audit trails, privacy and responsible deployment

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 5 NOTES · PART 1

AI/ML use in forensic triage, classification and anomaly detection

AI/ML use in forensic triage, classification and anomaly detection is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Supervised vs unsupervised learning for security evidence

Compare the alternatives in Supervised vs unsupervised learning for security evidence using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Behaviour analytics and detection of suspicious patterns

Behaviour analytics and detection of suspicious patterns is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on AI-Based Digital Forensics and Incident Response, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 5 NOTES · PART 2

Automated log correlation, alert prioritisation and threat intelligence

Automated log correlation, alert prioritisation and threat intelligence is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Benefits: speed, scale and pattern discovery

Benefits: speed, scale and pattern discovery is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on AI-Based Digital Forensics and Incident Response, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 5 NOTES · PART 3

Risks: false positives/negatives, bias, explainability and adversarial inputs

The OSI model has Physical, Data Link, Network, Transport, Session, Presentation and Application layers. Each layer provides services to the one above and uses the one below; encapsulation adds headers as data moves down.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Human validation, audit trails, privacy and responsible deployment

For Human validation, audit trails, privacy and responsible deployment, discuss the objective, stakeholders and possible harm. Cover consent/authority, confidentiality, integrity, accountability, transparency and applicable policy or law. Recommend safeguards and explain how they are checked.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on AI-Based Digital Forensics and Incident Response, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **AI/ML use in forensic triage, classification and anomaly detection** — know the definition, core working and one example.

• **Supervised vs unsupervised learning for security evidence** — know the definition, core working and one example.

• **Behaviour analytics and detection of suspicious patterns** — know the definition, core working and one example.

• **Automated log correlation, alert prioritisation and threat intelligence** — know the definition, core working and one example.

• **Benefits: speed, scale and pattern discovery** — know the definition, core working and one example.

• **Risks: false positives/negatives, bias, explainability and adversarial inputs** — know the definition, core working and one example.

• **Human validation, audit trails, privacy and responsible deployment** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from AI-Based Digital Forensics and Incident Response. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define AI/ML use in forensic triage, classification and anomaly detection and explain its importance.

Q2. Explain the working/steps involved in Supervised vs unsupervised learning for security evidence.

Q3. Differentiate between two related concepts from AI-Based Digital Forensics and Incident Response.

Q4. Give one practical application of Human validation, audit trails, privacy and responsible deployment and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of AI-Based Digital Forensics and Incident Response.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.